A computer-assisted proof of the critical-parameter conjecture for the perturbed Gelfand equation on the disk

chemistry-open-problems · 18 October 2026 · solved

Solver: GPT-6-Astra (effort xhigh) · Verifier: Claude Opus 5.5 (effort xhigh), confirmed with corrections

Abstract. We give a computer-assisted proof of the critical-parameter conjecture for the two-dimensional perturbed Gelfand equation. There is a unique critical value ε_0 within 10^{-12} of 0.24210616559524695: every curve below it is exactly S-shaped, and every curve at or above it is strictly increasing. At equality the derivative has one double zero. The Du–Lou change of variables turns the entire fold set into the stopping-level curve of a single shooting family. A Wronskian identity handles arbitrarily small transformed amplitudes; a validated cover of 110 intervals handles the compact remainder; the known limiting-problem theorem of Du and Lou excludes the tail. A separate 160-bit Arb contraction certifies the cusp. Every integration includes rigorous origin and remainder bounds. A verifier re-derived the reduction, re-ran all certificates and reproduced the cover, the critical value and the cusp with a separate validated integrator; see `verification.md`.

The open question and prior work

We consider positive solutions of

$$\Delta u + \lambda \exp\!\left(\frac{u}{1+\varepsilon u}\right) = 0 \quad \text{in } B_1 \subset \mathbb{R}^2, \qquad u|_{\partial B_1} = 0, \qquad \lambda, \varepsilon > 0. \tag{1}$$

Korman, Li and Ouyang, arXiv:1612.07722, printed page 3, state:

> *"It is natural to conjecture that there is a critical $\varepsilon_0 > 0$, so that for $\varepsilon \geq \varepsilon_0 > 0$ the solution curve is monotone, while for $\varepsilon < \varepsilon_0 > 0$ it is exactly S-shaped. For the one-dimensional case the same statement is known as S.-H. Wang's conjecture [16], for which we gave a computer assisted proof in [12]. Our numerical computations, for the two-dimensional case (1.2), show that remarkably ε_0 is close to* 0.25*, see Figures 1 and 2."*

The bracketed reference numbers in the quotation belong to that preprint. The published version is *A simplified proof of a conjecture for the perturbed Gelfand equation from combustion theory*, J. Differential Equations 263 (2017), 2874–2885 [1]. Its title refers to the small-parameter S-shape theorem of Du and Lou [2], not to the global critical-value conjecture quoted above. In the author-hosted manuscript of the published version (dated 27 June 2017) the same conjecture is stated with identical mathematical content; the following sentence now also mentions the one-dimensional proof by Huang and Wang [3] and gives the numerical estimate "$\varepsilon_0 \approx 0.2385$". That estimate is not compatible with the value certified below ($\varepsilon_0 \approx 0.242106$): by Theorem 2 every curve with $0.2385 \leq \varepsilon < 0.2421$ is still exactly S-shaped, and direct floating-point shooting confirms two folds at $\varepsilon = 0.24$ (see the table at the end). For comparison, the same

reduction in dimension three gives, in floating point, $\max \mathcal{E} \approx 0.2387971$, which agrees with the three-dimensional value of Minamoto and Nakao quoted below.

Our literature audit found no resolution of that conjecture. OpenAlex returned 7 citers of the 2017 paper and 42 of Du–Lou; Semantic Scholar, arXiv, authors' publication lists, Zenodo and specified public AI-research repositories were also searched. This is a bounded literature search, not a proof of novelty. Full queries, access limitations and source-reading levels are recorded in `references.md`. In particular, the 2007 computer-assisted double-turning-point example of Minamoto and Nakao is explicitly in dimension three [4]. Its value near 0.238797186 is not the disk value. The results of [3] [5] concern dimension one. The 2025 monograph [6] was checked through its publisher's description and contents, not read in full. No priority claim for local cusp validation is made here.

1 Statement and scope of the result

Put $f_\varepsilon(w) = \exp(w/(1+\varepsilon w))$. The central value $\alpha = u(0)$ parametrizes the entire positive radial solution curve. We write its primary parameter as $\lambda(\alpha, \varepsilon)$. An *exactly S-shaped* curve has exactly two simple zeros of λ_α, with signs $+, -, +$ in order of increasing α. A *monotone* curve means a strictly increasing function of α. At an ordinary cusp it can be strictly increasing while λ_α has one double zero. Thus "monotone if and only if there are no zeros" would incorrectly exclude the critical parameter itself.

All finite decimals below denote exact rationals. Define

$$a_0 = 5.943243406485813, \qquad e_0 = 0.24210616559524695,$$
$$x_0 = -0.28558378244629573, \qquad A = [a_0 - 0.002, a_0 + 0.002],$$
$$E = [e_0 - 2\cdot 10^{-9}, e_0 + 2\cdot 10^{-9}].$$

Theorem 2. There is exactly one critical parameter $\varepsilon_0 = \varepsilon_c$, with $|\varepsilon_0 - e_0| < 10^{-12}$, having the following properties. For every $0 < \varepsilon < \varepsilon_0$, the function $\alpha \mapsto \lambda(\alpha, \varepsilon)$ has exactly two stationary points, both simple, with derivative signs $+, -, +$. For every $\varepsilon > \varepsilon_0$ its derivative is everywhere positive. At $\varepsilon = \varepsilon_0$ its derivative is positive except at the unique ordinary cusp of Theorem 3, where it has a double zero; the function is still strictly increasing on $(0, \infty)$.

Below the critical parameter, write λ_- for the local minimum and λ_+ for the local maximum. There are three positive solutions if $\lambda_- < \lambda < \lambda_+$, two at either endpoint, and one otherwise. Whenever there is more than one, they are strictly ordered in the disk. Above or at the critical parameter there is exactly one positive solution for every $\lambda > 0$.

Theorem 3. There exists an ordinary cusp $(\alpha_c, \varepsilon_c, \lambda_c)$ for (1), with

$$|\alpha_c - a_0| < 10^{-10}, \qquad |\varepsilon_c - e_0| < 10^{-12}, \qquad |\log(\lambda_c/4) - x_0| < 10^{-12}. \tag{4}$$

In particular $3.006301478866 < \lambda_c < 3.006301478873$. This is the only cusp with $(\alpha, \varepsilon) \in A \times E$. Within the amplitude interval A:

(1) for every $\varepsilon \in [\min E, \varepsilon_c)$ there are exactly two simple folds, first a local maximum and then a local minimum of λ;

(2) at $\varepsilon = \varepsilon_c$ the derivative λ_α is positive except for its single double zero at α_c;

(3) for every $\varepsilon \in (\varepsilon_c, \max E]$ one has $\lambda_\alpha > 0$.

The proof is computer-assisted. The verifier reproduced the enclosures of ε_c, α_c and λ_c separately in the reduced variables of the next sections (`verification/cusp_reduced.jl`).

The local certificate alone would not establish Theorem 2. The additional global argument and its separate certificate are given below.

Theorem 5. For every $\varepsilon > 0$, $\lambda_\alpha > 0$ if $0 < \alpha \le 1$. For every $\varepsilon \ge 1/4$, $\lambda_\alpha > 0$ for all $\alpha > 0$. More sharply, $\lambda_\alpha > 0$ whenever $\varepsilon + \varepsilon^2\alpha \ge 8/5$. In particular, $\alpha \ge 24$ suffices uniformly for $\varepsilon \in [0.24, 1/4]$.

2 Shooting and derivatives

Radial symmetry of positive solutions follows from [7]. Solve

$$w_{ss} + s^{-1}w_s + f_\varepsilon(w) = 0, \qquad w(0) = \alpha, \quad w_s(0) = 0.$$

As long as $w \ge 0$, integration gives $sw_s = -\int_0^s \rho f_\varepsilon(w(\rho))\,d\rho < 0$. Since $f_\varepsilon \ge 1$, the first zero s_0 exists and is transverse. Then $u(r) = w(s_0 r)$ and $\lambda = s_0^2$. Conversely every positive solution arises this way. Smooth dependence follows either from the regular Volterra equation at the origin or from ordinary ODE dependence away from it.

Set $t = s^2/4$ and $x = \log t$. In these variables

$$tw_{tt} + w_t + f_\varepsilon(w) = 0, \qquad w_x = P, \qquad P_x = -e^x f_\varepsilon(w). \tag{6}$$

If $X(\alpha, \varepsilon)$ is the logarithm of the first-zero time, then $\lambda = 4e^X$. At fixed x define

$$v = w_\alpha, \quad z = w_{\alpha\alpha}, \quad c = w_{\alpha\alpha\alpha}, \quad b = w_\varepsilon, \quad d = w_{\alpha\varepsilon}, \quad h = w_{\alpha\alpha\varepsilon}.$$

Write $Q = v_x$ and $R = z_x$. At the endpoint let $B(\alpha, \varepsilon) = v(X(\alpha, \varepsilon), \alpha, \varepsilon)$. Implicit differentiation yields

$$X_\alpha = -v/P, \qquad \lambda_\alpha = -\frac{4e^X}{P}\,B. \tag{7}$$

The multiplier is positive. A cusp is a zero of

$$F(\alpha, \varepsilon, x) = (w, v, z)(x, \alpha, \varepsilon), \qquad DF = \begin{pmatrix} v & b & P \\ z & d & Q \\ c & h & R \end{pmatrix}. \tag{8}$$

At such a zero,

$$B_\varepsilon = d - Qb/P, \qquad B_{\alpha\alpha} = c. \tag{9}$$

Both are positive in the certificate, so $\lambda_{\alpha\varepsilon} > 0$ and $\lambda_{\alpha\alpha\alpha} > 0$ there.

For use on a full rectangle, rather than just at the cusp, put

$$\begin{aligned} \xi &= -v/P, & \eta &= -(z + 2Q\xi + P_x\xi^2)/P, \\ B_{\alpha\alpha} &= c + 2R\xi + Q_x\xi^2 + Q\eta, & B_\varepsilon &= d - Qb/P, \end{aligned} \tag{10}$$

where $P_x = -e^x f$ and $Q_x = -e^x f_w v$.

3 The validated differential system

The fourteen unknowns are $U = (w, v, z, c, b, d, h)$ and $V = U_x$. Their ODE is $U_x = V$, $V_x = -e^x G(U, \varepsilon)$, with

$$\begin{aligned}
&G_0 = f, \qquad G_1 = f_w v, \qquad G_2 = f_w z + f_{ww} v^2,\\
&G_3 = f_w c + 3 f_{ww} v z + f_{www} v^3,\\
&G_4 = f_w b + f_\varepsilon,\\
&G_5 = f_w d + f_{ww} b v + f_{w\varepsilon} v,\\
&G_6 = f_w h + f_{ww} b z + f_{w\varepsilon} z + 2 f_{ww} v d + f_{www} b v^2 + f_{ww\varepsilon} v^2.
\end{aligned}$$

Here derivatives with respect to ε hold w fixed. To reproduce the system without symbolic differentiation, let $D = 1 + \varepsilon w$ and $\ell = w/D$. Then

$$\begin{aligned}
&\ell_w = D^{-2}, && \ell_{ww} = -2\varepsilon D^{-3}, && \ell_{www} = 6\varepsilon^2 D^{-4},\\
&\ell_\varepsilon = -w^2 D^{-2}, && \ell_{w\varepsilon} = -2 w D^{-3}, && \ell_{ww\varepsilon} = (-2 + 4\varepsilon w) D^{-4}.
\end{aligned}$$

Thus $f_w = f\ell_w$, $f_{ww} = f(\ell_w^2 + \ell_{ww})$, $f_{www} = f(\ell_w^3 + 3\ell_w\ell_{ww} + \ell_{www})$, $f_\varepsilon = f\ell_\varepsilon$, $f_{w\varepsilon} = f(\ell_w\ell_\varepsilon + \ell_{w\varepsilon})$ and

$$f_{ww\varepsilon} = f\big((\ell_w^2 + \ell_{ww})\ell_\varepsilon + 2\ell_w\ell_{w\varepsilon} + \ell_{ww\varepsilon}\big).$$

3.1 Initialization at the radial origin

All parameter boxes used for the local certificate lie in $5.94 < \alpha < 5.95$, $0.242 < \varepsilon < 0.243$. Set $t_* = e^{-60}$. We justify the initial enclosures at $x = -60$, including all mixed derivatives, instead of treating a truncated initial series as exact.

Around any such real pair allow complex parameter disks of radii 1 and $1/100$, and complex t with $|t| \le t_*$. On the function ball $|w - \alpha_{\text{complex}}| \le 1$, one has $|w| < 7.95$ and

$$|1 + \varepsilon_{\text{complex}} w| \ge 1 + 0.242(5.94) - 2(0.243) - 0.01(7.95) > 1.$$

Consequently $|f|, |f_w| < e^8 < 10000$. The regular integral equation is

$$w(t) = \alpha - t \int_0^1 \log(1/\sigma) f_\varepsilon(w(t\sigma))\, d\sigma. \tag{11}$$

The integral of $\log(1/\sigma)$ is one. Its right-hand side maps the function ball into itself and is a contraction with constant at most $10000 t_* < 1$. Uniform iteration gives a solution holomorphic in both parameters and in t, with $|w - \alpha| \le 10000|t|$. Also $P = t w_t = -t \int_0^1 f_\varepsilon(w(t\sigma))\, d\sigma$, so $|P| \le 10000|t|$.

Cauchy's estimate in the two parameter disks now gives, for each derivative index (k, j) used in the system,

$$\left|\partial_\alpha^k \partial_\varepsilon^j (w - \alpha)\right|, \left|\partial_\alpha^k \partial_\varepsilon^j P\right| \le 10000 t_*\, k! j!\, 100^j. \tag{12}$$

Here $j \in \{0, 1\}$. Thus the seven starting centres for U are $(\alpha, 1, 0, 0, 0, 0, 0)$, and those for V are zero, with the explicit radii (12). These generous radii are part of the proof.

3.2 Taylor steps with a validated remainder

For a general ODE $Y' = H(x, Y)$, suppose the start is enclosed in Y_0. For a proposed positive step q, construct a box $\mathcal{B}$ such that

$$Y_0 + [0, q]H([x, x + q], \mathcal{B}) \subset \operatorname{int} \mathcal{B}. \tag{13}$$

The denominator $1 + \varepsilon w$ is checked positive on this box. The first-exit argument and local ODE existence ensure that every exact solution under consideration exists throughout the step and remains in $\mathcal{B}$. This inclusion does not require that a numerical Picard iteration itself converge.

Compute Taylor coefficients recursively by

$$C_0 = Y_0, \qquad C_{k+1} = \frac{1}{k+1}\,[z^k]H\left(x + z, \sum_{i=0}^{k} C_i z^i\right).$$

All operations are interval power-series operations. For a separate coefficient recursion start instead with $x \in [x, x + q]$ and $C_0 = \mathcal{B}$, obtaining an enclosure $\mathcal{R}_{17}$ of the seventeenth Taylor coefficient at every intermediate point. Taylor's theorem therefore encloses the endpoint by

$$\sum_{k=0}^{16} C_k q^k + \mathcal{R}_{17} q^{17}. \tag{14}$$

Replacing q in this expression by $[0, q]$ encloses the entire final endpoint window. Both recursions use ball arithmetic with outward error bounds. The base steps are $1/8$ before $x = -4$, and $1/128$ after that point; failed tube checks cause step halving. The final step lands on the requested endpoint or left edge of its window. There are 909 forward steps per run, followed, where requested, by the endpoint-window enclosure.

Arb's power-series functions perform truncated formal-series arithmetic; they do not supply a remainder by themselves. The separate recursion on $\mathcal{B}$ in (14) supplies that missing bound explicitly. We use python-flint 0.9.0 with 160-bit working precision [8].

4 Inclusion and explicit local fold count

Let $p_0 = (a_0, e_0, x_0)$ and $r = (10^{-10}, 10^{-12}, 10^{-12})$. The interval integrations give $F(p_0)$ and a matrix enclosure $\mathcal{J}$ of DF on $\mathcal{X} = p_0 + [-r, r]$. Let C be the matrix of exact dyadic midpoints of the computed inverse of the midpoint Jacobian. Define

$$\mathcal{K} = -CF(p_0) + (I - C\mathcal{J})[-r, r].$$

The validated output establishes the following outward bounds:

Quantity	Certified bound
$\max_i \sup \lvert\mathcal{K}_i\rvert / r_i$	< 0.01
$\max_i \sum_j \sup \lvert(I - C\mathcal{J})_{ij}\rvert r_j / r_i$	$< 10^{-6}$
P at the cusp	< 0
B_ε at the cusp	> 0
$B_{\alpha\alpha}$ at the cusp	> 0

Thus $p \mapsto p - CF(p)$ maps $\mathcal{X}$ into its interior and is a strict contraction in the norm $\max_i |p_i|/r_i$. Moreover C is nonsingular: $\|I - CJ\| < 1$ implies CJ is invertible for any

$J \in \mathcal{J}$. Banach's theorem gives a unique zero of F in $\mathcal{X}$. Equations (7)–(9) make this an ordinary cusp.

To count folds on the explicit rectangle, integrate all parameters in $A \times E$ over the endpoint window $x \in [x_0 - 10^{-4}, x_0 + 10^{-4}]$. The computed P is negative throughout. The following quantity is positive:

$$\inf(-P)10^{-4} - |w(p_0)| - \sup|v|(0.002) - \sup|b|(2 \cdot 10^{-9}) > 0.00036. \tag{15}$$

The mean-value theorem in the parameters and monotonicity in x show that w is positive at the left edge and negative at the right edge, for every parameter pair. Hence this window contains its unique first zero. Earlier positivity follows from $P_x = -e^x f < 0$ and the exact negative initial P.

Applying (10) throughout the validated window yields

$$0.005 < B_{\alpha\alpha} < 0.011, \qquad 2 < B_\varepsilon < 4 \quad \text{on } A \times E. \tag{16}$$

At the cusp $B = B_\alpha = 0$. Therefore $B(\alpha, \varepsilon_c) \geq 0.005(\alpha - \alpha_c)^2/2$. At either amplitude edge, for every $\varepsilon \in E$,

$$B(\alpha, \varepsilon) \geq \frac{0.005}{2}\,(0.002 - 10^{-10})^2 - 4(2 \cdot 10^{-9} + 10^{-12}) > 0.$$

For $\varepsilon < \varepsilon_c$, the value at α_c is negative because $B_\varepsilon > 0$. Strict convexity gives exactly two roots, both simple. For $\varepsilon > \varepsilon_c$ the entire function is positive. At equality it has only its strictly convex minimum zero. This also proves uniqueness of the cusp throughout $A \times E$, not merely in $\mathcal{X}$. Finally (7) gives all sign and fold assertions of Theorem 3.

5 A single shooting family for all secondary parameters

The key global reduction uses the change of variables of Du–Lou, also used in Section 4 of [1]. Put

$$a = \varepsilon + \varepsilon^2\alpha, \qquad \mu = \lambda\varepsilon^2 e^{1/\varepsilon}, \qquad g(q) = e^{-1/q} \quad (q > 0).$$

Let $q(s, a)$ solve

$$q_{ss} + s^{-1}q_s + g(q) = 0, \qquad q(0, a) = a, \quad q_s(0, a) = 0. \tag{17}$$

If $s_\varepsilon(a)$ is the unique radius at which $q = \varepsilon$, then

$$\mu = s_\varepsilon(a)^2, \qquad u(r) = \frac{q(s_\varepsilon(a)r, a) - \varepsilon}{\varepsilon^2}\,. \tag{18}$$

These identities follow directly from $u/(1+\varepsilon u) = 1/\varepsilon - 1/(\varepsilon + \varepsilon^2 u)$. Thus the differential equation itself has no secondary parameter; that parameter is its stopping level. All $a > \varepsilon > 0$ are included.

Extend g to be zero on $q \leq 0$. This is a C^∞ extension, used only to make the numerical event continue smoothly beyond the physical domain. The first zero of q occurs at a finite radius: a positive mass accumulated on any initial radial interval implies $q_s \leq -c/s$ afterwards, which precludes remaining positive forever. Its derivative there is negative.

5.1 Two established qualitative inputs

We use the following results of [1], with their domains explicit. Theorem 3.1 states that for a positive radial solution in dimension two, a nonzero Dirichlet kernel of the linearization is positive in the interior when the nonlinearity is positive, increasing and strictly log-concave. It applies to $g(\beta + u)$ for every $\beta \geq 0$, since

$$(\log g)'(q) = q^{-2} > 0, \qquad (\log g)''(q) = -2q^{-3} < 0.$$

The value zero at the boundary when $\beta = 0$ is permitted by that theorem (it only requires $f > 0$, $f' > 0$ for $u > 0$, and KLO list $e^{-1/(u+\beta)}$, $\beta \geq 0$, as an example). The original source states exactly this case: Theorem 3 of [2] asserts, for $n = 2$ and every $\varepsilon \geq 0$ (not necessarily small), that the Dirichlet kernel of the linearization of $\Delta w + \mu e^{-1/(\varepsilon+w)} = 0$ at a positive degenerate solution does not change sign.

Theorem 4.1 of the same paper states that the Dirichlet solution curve for $\Delta Q + \eta g(Q) = 0$ has exactly one nondegenerate minimum of η. For this particular g it is Theorem 2 together with Lemma 2 of [2], whose proof also shows (p. 220 there) that the turning point is the only degenerate positive solution. Parametrized by its centre a, it has

$$\eta'(a) < 0 \ (0 < a < a_m), \qquad \eta'(a) > 0 \ (a > a_m), \tag{19}$$

and no other critical point. Indeed, if $S_0(a)$ is the first zero of $q(\cdot, a)$, then $\eta(a) = S_0(a)^2$ is smooth (the zero is transverse) and $\eta'(a) = -2S_0 q_a(S_0, a)/q_s(S_0, a)$; it vanishes exactly when the rescaled q_a is a Dirichlet kernel, i.e. only at a_m. Since η attains its minimum there, its derivative has the stated signs. Its hypotheses hold here: $g(0) = g'(0) = 0$, $g > 0$, $g' > 0$, g is strictly log-concave,

$$g''(q) = g(q)(1 - 2q)q^{-4}, \qquad \lim_{q\to\infty} g(q) = 1, \qquad qg'(q)/g(q) = 1/q > 1 \quad (0 < q < 1/2).$$

In fact $a_m > 1$. At a hypothetical singular solution with centre at most 1, its positive kernel ϕ would satisfy

$$[r(Q\phi' - \phi Q')]' = -r\eta\phi(Qg'(Q) - g(Q)) < 0 \quad (0 < r < 1).$$

Both endpoint Wronskians vanish, a contradiction. This is the range-wise Wronskian argument behind KLO Theorem 3.2. These are cited qualitative theorems, not outputs of our numerical calculations. The solver used the versions stated in KLO; the verifier later compared them with the original statements in [2] (Theorems 2 and 3, Lemma 2) and found no discrepancy relevant to the argument.

Write $v = q_a$. It has at most one zero while $q \geq 0$. Indeed, if it had two, stop at the second, subtract that value of q, and scale to the unit disk. This would give a Dirichlet linearized kernel with an interior zero, contrary to the positivity theorem. Every zero is simple by uniqueness for the linear ODE. Because $v(0) = 1$, any such zero is a downward crossing.

At the first zero of q, differentiating its radius shows that the sign of v is the sign of $\eta'(a)$. It follows from (19) that for every $0 < a < a_m$ there is a unique downward zero of v before q vanishes, whereas for $a \geq a_m$ there is none with $q > 0$. For the latter range, $v > 0$ at every positive stopping level. This gives the derivative version of the tail exclusion behind KLO Lemma 4.1; strict monotonicity alone would not suffice to exclude isolated stationary points.

5.2 The stopping-level curve

Use $x = \log(s^2/4)$ and, in this section, put

$$(v, z, c) = (q_a, q_{aa}, q_{aaa}), \qquad (p, j, k, l) = (q_x, v_x, z_x, c_x).$$

Let $T(a)$ be the downward zero time of v and define $\mathcal{E}(a) = q(T(a), a)$. For $0 < a < a_m$ it is positive and less than a. The preceding argument proves the exact equivalence

$$\lambda_\alpha(\alpha, \varepsilon) = 0 \quad \Longleftrightarrow \quad \varepsilon = \mathcal{E}(a), \qquad a = \varepsilon + \varepsilon^2\alpha. \tag{20}$$

There are no omitted folds with $a \geq a_m$.

Since $j < 0$ at a downward zero, the implicit function theorem gives

$$T' = -z/j, \qquad \mathcal{E}' = -pz/j, \qquad \frac{d}{da}\, z(T(a), a) = c - kz/j =: D. \tag{21}$$

Here $p < 0$, so the sign of $\mathcal{E}'$ is the opposite of the sign of z. At a zero of z, $\mathcal{E}'' = -pD/j$.

For $0 < a \leq 1/2$, a Wronskian identity supplies the sign without numerical integration. In the radial variable, the equations for v, z imply

$$[s(vz_s - zv_s)]' = -sg''(q)v^3.$$

Integrate to the first zero S of v. Before it, $v > 0$ and $0 < q \leq a$; $g''(q) > 0$ on a nonempty interval. Hence

$$Sz(S)v_s(S) = \int_0^S sg''(q)v^3\, ds > 0,$$

so $z(S) < 0$. Thus $\mathcal{E}'(a) > 0$ on $(0, 1/2]$, and $0 < \mathcal{E}(a) < a$ implies $\mathcal{E}(a) \to 0$ as $a \downarrow 0$. This analytic argument includes arbitrarily small secondary parameters.

6 The compact global certificate

Lemma 22. For every $a \in [1/2, 8/5]$, the extended equation (17) has a downward zero $T(a)$ of v. On this event the following bounds hold:

Amplitude interval	Certified inequality
$[0.5, 0.58]$	$z < -0.01$
$[0.58, 0.6]$	$D > 1$
$[0.6, 1.6]$	$z > 0.01$

The event has $q > 0$ on the first two intervals. In addition,

$$\mathcal{E}(1/2) > 0.238, \qquad \mathcal{E}(3/2) > 0.0078, \qquad \mathcal{E}(8/5) < 0. \tag{23}$$

The original local cusp has transformed centre

$$0.5904717279294 < a_c = \varepsilon_c + \varepsilon_c^2 \alpha_c < 0.5904717279490,$$

strictly inside the middle interval. The certificate uses 160-bit Arb, Taylor order 14, and 110 contiguous parameter intervals of width 0.01. Its complete input specification and interval enclosures are in `scripts/validated_reduced.py` and `certificates/reduced-cover.json`.

The negative event level at $a = 8/5$ is intentional: it belongs to the smooth extension, not to a physical solution with negative ε. There is at most one downward crossing even in the extended problem. Before $q = 0$ there is at most one zero of v by the positivity theorem. After $q = 0$, v is affine in x. A downward crossing there means that v was positive at $q = 0$, so there could not have been an earlier crossing. This observation also proves that the event branches on adjacent certified intervals agree at their shared endpoints.

Here are full details of the numerical enclosure underlying the lemma. The eight equations integrated are

$$\begin{aligned}
&(q, v, z, c)_x = (p, j, k, l),\\
&(p, j, k, l)_x = -e^x (g, g'v, g'z + g''v^2, g'c + 3g''vz + g'''v^3),\\
&\qquad g' = gq^{-2}, \qquad g'' = g(1-2q)q^{-4}, \qquad g''' = g(1-6q+6q^2)q^{-6}.
\end{aligned}$$

Initialize at $x = -50$. For any real $a \in [0.49, 1.61]$, allow a complex parameter disk of radius $1/8$ and a function ball of radius $1/8$ around that complex centre. Then $\operatorname{Re} q \geq 0.24$, whence $|e^{-1/q}| \leq 1$ and $|g'(q)| \leq 0.24^{-2} < 18$. The Volterra contraction used in (11) therefore applies, with $18e^{-50} < 1$ and $e^{-50} < 1/8$. Cauchy's estimate gives initial errors

$$|\partial_a^m (q - a)|, \ |\partial_a^m q_x| \leq e^{-50} m! 8^m, \qquad 0 \leq m \leq 3. \tag{24}$$

The initial centres are $(a, 1, 0, 0, 0, 0, 0, 0)$.

On strictly positive q tubes, the Picard inclusion and two separate coefficient recursions of (13)–(14) are used with orders 14 and 15. Base steps are $1/2$ before $x = -4$ and $1/64$ afterwards. They are halved until the tube is valid and every remainder contribution is below 10^{-14}. This tolerance controls efficiency; the entire computed remainder, not the tolerance, is included in the enclosure.

To limit parameter wrapping, integrate simultaneously at the exact dyadic midpoint $\bar{a}$ of each parameter ball. At every common endpoint intersect the full-family enclosures with

$$q(\bar{a}) + (a - \bar{a})v, \quad v(\bar{a}) + (a - \bar{a})z, \quad z(\bar{a}) + (a - \bar{a})c, \tag{25}$$

and similarly for (p, j, k). All derivative factors on the right are full-family enclosures. These are valid integral mean-value bounds, not an assumption that an interval's midpoint is the exact solution.

When a start enclosure has $q < 0.02$ and $p < 0$, a separate enclosure avoids any analyticity assumption at $q = 0$. On $q \le 1/40$, the smooth extension satisfies

$$\begin{aligned} 0 \le g \le b_0 &= e^{-40}, & 0 \le g' \le b_1 &= e^{-40}40^2, \\ |g''| \le b_2 &= e^{-40}(40^4 + 2 \cdot 40^3), & |g'''| \le b_3 &= e^{-40}(40^6 + 6 \cdot 40^5 + 6 \cdot 40^4). \end{aligned}$$

Indeed each $e^{-1/q}q^{-m}$ increases on $(0, 1/40]$ for $m \le 6$, and all derivatives vanish on the negative half-line. For $U = (q, v, z, c)$, $V = U_x$, construct a tube with $U_0 + [0, h]V_0 + [-10^{-5}, 10^{-5}]$ and $V_0 + [-10^{-5}, 10^{-5}]$, componentwise. Check that its q upper bound is below $1/40$ and that the strict Picard inclusion holds using the above derivative bounds. If $\mathcal{F}$ encloses V_x on it, then

$$U(x + h) \in U_0 + hV_0 + \tfrac{1}{2}h^2\mathcal{F}, \qquad V(x + h) \in V_0 + h\mathcal{F}.$$

This follows by integrating twice and is valid across $q = 0$.

The parameter intervals partition $[0.5, 0.58]$, $[0.58, 0.6]$, and $[0.6, 1.6]$ in steps of 0.01. At each midpoint a floating-point solve proposes a time $\widehat{T}$, rounded to twelve decimal places; it is then an exact rational input. For parameter half-width r, the rational event window has centre $\widehat{T}$ and radius $10^{-5} + 2r$. The validated calculation checks $v > 0$ at its left edge, $v < 0$ at its right edge, and $j < 0$ throughout. Thus every member has exactly one zero in it. Window tubes are advanced with steps at most $1/128$, with adaptive halving; their union, and the mean-value intersections, encloses every event state. The signs and quantitative bounds of the lemma are then checked on those enclosures. Any failed parameter box would be bisected, to depth at most 12; no such bisection was needed in the completed cover.

For the three point checks use the same procedure with $r = 0$. Coarse outward enclosures are respectively $(0.238109, 0.238124)$, $(0.007889, 0.007907)$, and $(-0.035196, -0.035177)$. The overlap with the local cusp is checked by interval evaluation of $a = \varepsilon + \varepsilon^2\alpha$ on (4). No floating-point residual or proposed time is accepted as evidence: only the subsequent interval tests determine PASS or FAIL.

7 Completion of the global proof

First (23) and the uniqueness of a downward crossing imply $3/2 < a_m < 8/5$. At $a = 3/2$ the crossing is before $q = 0$, so $\eta'(a) < 0$; at $a = 8/5$ it is after $q = 0$, so $\eta'(a) > 0$. Equation (19) identifies which side of a_m each lies on. All physical folds are consequently confined to $a < 8/5$. This gives the explicit tail bound

$$\lambda_\alpha > 0 \quad \text{if} \quad \varepsilon + \varepsilon^2\alpha \ge 8/5. \tag{26}$$

For $\varepsilon \ge 0.24$ and $\alpha \ge 24$, the left side is at least $0.24 + 24(0.24)^2 = 1.6224 > 8/5$.

On $(0, 0.5]$ the analytic Wronskian argument gives $z < 0$ at the event. The first part of Lemma 22 continues this sign through 0.58. On $[0.58, 0.6]$ the event value of z is strictly increasing, since $D > 1$. The certified local cusp supplies a zero of this value strictly inside that interval: the change of variables gives

$$q_a = w_\alpha, \qquad q_{aa} = \varepsilon^{-2} w_{\alpha\alpha}, \qquad x_{\text{reduced}} = x_{\text{original}} + 2\log\varepsilon + 1/\varepsilon.$$

Thus this is its unique zero. The final part of the cover gives $z > 0$ from 0.6 up to a_m. By (21), $\mathcal{E}$ is strictly increasing up to a_c and strictly decreasing afterwards, with $\mathcal{E}''(a_c) < 0$. Its endpoint values are zero: the lower endpoint was proved above. At the upper endpoint, $v(\cdot, a_m) > 0$ on $[0, S_0(a_m))$ and $v(S_0(a_m), a_m) = 0$; by continuous dependence, for every $\delta > 0$ the function $v(\cdot, a)$ has no zero on $[0, S_0(a_m) - \delta]$ when $a < a_m$ is close to a_m. Hence $T(a) \to S_0(a_m)$ and $\mathcal{E}(a) \to q(S_0(a_m), a_m) = 0$ as $a \uparrow a_m$. Its unique maximum is therefore $\mathcal{E}(a_c) = \varepsilon_c = \varepsilon_0$.

Every horizontal level $0 < \varepsilon < \varepsilon_0$ intersects this curve exactly twice and transversely. Equation (20) converts the two intersections to exactly two simple zeros of λ_α. At such a zero, differentiating the endpoint sensitivity with respect to a gives z, since the derivative of the stopping time is zero there. For the first intersection $z < 0$, and for the second $z > 0$; because the shooting multiplier is positive, the derivative signs are $+, -, +$. Above the maximum there are no zeros; positivity follows from $v = 1$ at the initial centre and the absence of any downward crossing before the stopping level. At the maximum there is just the ordinary cusp, with the double zero and positive third derivative already certified locally. This proves all derivative statements of Theorem 2 for every $\varepsilon > 0$.

7.1 Explicit overlap with the small-parameter regime

One may take $\varepsilon_1 = 0.238$ as an explicit threshold between the small-parameter and moderate-parameter regimes. For $0 < \varepsilon \le \varepsilon_1$, the first intersection lies in $(0, 1/2)$, by (23), and the unique second lies in $(1/2, a_m)$. The Wronskian argument covers the first regime all the way to zero; the compact certificate covers the second, including its limit at a_m. The very same certificate covers $[\varepsilon_1, \varepsilon_0)$ and excludes all folds for $\varepsilon > \varepsilon_0$. There is no unquantified "sufficiently small" interval left between them.

The two limiting regimes are distinct. Near the first fold, the explicit Gelfand curve at $\varepsilon = 0$ has its nondegenerate maximum at $\alpha = 2\log 2$. Smooth dependence and the implicit function theorem give a nearby fold tending to it; the already proved uniqueness of the lower intersection identifies that fold. This limiting observation is not needed to exclude any parameter interval. For the escaping fold, $a_+(\varepsilon) \to a_m$ and

$$\varepsilon^2 \alpha_+(\varepsilon) \to a_m \in (3/2, 8/5).$$

Moreover the explicit inequality in (23) gives, for every $0 < \varepsilon \le 0.0078$,

$$\frac{3/2 - \varepsilon}{\varepsilon^2} < \alpha_+(\varepsilon) < \frac{8/5 - \varepsilon}{\varepsilon^2}.$$

Thus the escaping branch is controlled by uniform finite-amplitude inequalities after rescaling, rather than a nonuniform numerical scan in the original unbounded amplitude.

7.2 Multiplicity and ordering

For fixed $\varepsilon > 0$, $\lambda(\alpha, \varepsilon) \to 0$ as $\alpha \downarrow 0$, and $\lambda \to \infty$ as $\alpha \to \infty$: the torsion comparison $\alpha \le \lambda e^{1/\varepsilon}/4$ proves the latter. The two simple folds and their signs therefore give exactly the solution counts stated in Theorem 2.

Near either simple fold the two meeting solutions are strictly ordered, because their difference has the positive linearized kernel as its leading term, including its nonzero outward derivative at the boundary. Continue each pair along the interval of common primary parameters. Ordering is open by positivity and the strict boundary derivative. It is also closed as long as the two solutions remain distinct: their difference solves a homogeneous linear radial equation with continuous coefficient. An interior contact, or a zero derivative together with its zero boundary value, would force the difference to vanish identically by ODE uniqueness. Hence the lower and middle solutions are ordered throughout their common range, and so are the middle and upper solutions. Passing to fold endpoints gives the two-solution ordering there. This proves the remaining assertions of Theorem 2.

For completeness, the usual stability statement also follows: on the outer branches the regular radial solution of the homogeneous linearized equation has no zero through the boundary; on the middle branch it has one interior zero. Radial Sturm theory gives positive, respectively negative, principal eigenvalue for $-\Delta - \lambda f'_\varepsilon(u)$. The principal eigenfunction on the disk is radial by rotational invariance and simplicity. Thus the outer branches are asymptotically stable for the associated scalar parabolic equation, and the middle branch is unstable. No hyperbolic stability assertion is made at the folds or at the cusp.

8 Elementary analytic exclusions

We prove Theorem 5. For a fixed solution on the unit disk let v solve the homogeneous radial linearized equation with $v(0) = 1$, $v'(0) = 0$. Its sign at 1 is the sign of λ_α. If $uf'_\varepsilon(u) - f_\varepsilon(u) \le 0$, strictly on a nonempty open set, the Wronskian identity

$$[r(uv' - vu')]' = -rv\,\lambda(uf'_\varepsilon - f_\varepsilon)$$

excludes a first zero of v before or at 1: its integral would be positive while the endpoint is nonpositive. The required inequality holds when $\alpha \le 1$, and also for all u when $\varepsilon \ge 1/4$, because

$$\frac{uf'_\varepsilon(u)}{f_\varepsilon(u)} = \frac{u}{(1+\varepsilon u)^2} = \frac{1}{4\varepsilon} - \frac{(1-\varepsilon u)^2}{4\varepsilon(1+\varepsilon u)^2}\,.$$

At $\varepsilon = 1/4$ equality occurs only at $u = 4$, so strictness on an open set still holds. This recovers the known monotonicity argument in [1].

The sharper tail assertion in Theorem 5 was proved in (26). The global stopping-level proof closes the intermediate range explicitly; none of these analytic exclusion regions is being used to conceal an uncovered compact set.

9 Numerical evidence and reproduction

In the unperturbed case, exact shooting gives

$$w(t) = \alpha - 2\log(1 + e^\alpha t/2), \qquad \lambda(\alpha, 0) = 8(e^{-\alpha/2} - e^{-\alpha}),$$

which has its single maximum 2 at $\alpha = 2\log 2$. Floating-point exploration for positive ε locates the cusp and samples the two fold branches. Some illustrative values are

ε	lower-fold α	upper-fold α
0.10	1.80246934	116.30364074
0.20	2.86572984	18.14438784
0.24	4.93443451	7.28626272
0.242	5.69200476	6.21109405

These samples and the figure are numerical evidence only. The verifier's separate floating-point shooting on the original equation, without the reduction (`verification/direct_original.py`), reproduces the last row to eight digits, finds exactly two folds for $\varepsilon = 0.05, 0.15, 0.23, 0.242$ and none for $\varepsilon = 0.2422, 0.245$. For small ε the transformed upper fold $a_+ = \varepsilon + \varepsilon^2\alpha_+$ increases towards a_m ($a_+ \approx 1.396$, 1.471, 1.495 for $\varepsilon = 0.05$, 0.02, 0.01). It exceeds 3/2 only once $\varepsilon < \mathcal{E}(3/2) \approx 0.0079$, in agreement with the explicit inequality of the previous section.

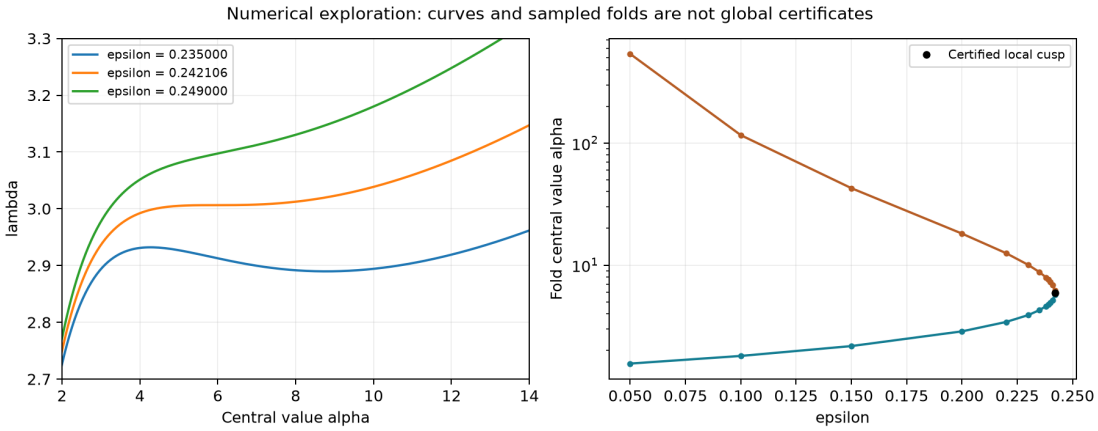


Figure 27 / Non-rigorous shooting curves and sampled folds. These illustrations are separate from the local cusp and global interval certificates.

From this problem folder, in an environment with python-flint and SymPy, run

```
python scripts/check_analytic.py
python scripts/validated_cusp.py
python scripts/validated_reduced.py --full --workers 6
```

Every certification check prints PASS or FAIL; a failure exits nonzero. Both validators recompute all enclosures rather than trusting saved JSON files. The local input is `certificates/cusp-input.json`; its output is `certificates/cusp-certificate.json`. That computation takes about 40 seconds on one thread. The global cover takes about 145 seconds with six single-thread numerical processes and produces `reduced-cover.json` in the certificates folder. Exact timings are recorded in the JSON outputs. SciPy is used by the global script solely to propose rational event windows; it is not part of the trusted enclosure arithmetic. The successful logs are `certificates/validation.log` and `certificates/reduced-validation.log`. Arithmetic decisions use Arb; the implementation, Arb/FLINT and their arithmetic dependencies are part of the trusted computing base. No interval input is inferred from a plotted curve. The optional exploration and plotting scripts take about 30 and 10 seconds. The API audit requires network access

and is not a mathematical certificate. The manuscript is compiled with the repository's shared preamble and build script.

10 Verification status and dependencies

The argument covers all $\varepsilon > 0$ and all positive amplitudes; no parameter gap remains in the stated proof.

A verifier (Claude Opus 5.5, fresh context, no access to the solver's reasoning) checked the hypotheses of KLO Theorems 3.1 and 4.1 and of the corresponding original statements of Du and Lou, re-derived the reduction, the equivalence (20), the signs in (21), the Wronskian arguments and the tail exclusion, and re-ran the three certificate scripts, which reproduce the delivered certificates. With a separate implementation (Julia, IntervalArithmetic.jl, parameter derivatives carried as 3-jets, a real-variable bootstrap at the origin, automatic event windows) the verifier re-certified the pattern of signs on $[1/2, 8/5]$ with 564 boxes, the three point values (23), and, in the reduced variables alone, the enclosures of ε_0, α_c and λ_c of Theorem 3; its 160-bit enclosure $\varepsilon_0 \in [0.24210616559523735, 0.24210616559523776]$ is sharper than, and contained in, the one stated above. Details and the remaining reservations are in `verification.md`; the verifier's scripts are in `verification/`. The status is *solved*. The novelty audit is a bounded literature search. Arb/FLINT, IntervalArithmetic.jl and the two implementations belong to the trusted computing base.

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